

Regression Week I: Foundations for linear models

Slide deck III: The normal distribution and related theorems

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Why the normal distribution matters

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- ▶ This assumption is not needed for ordinary least squares point estimation.
- ▶ It is central for the exact finite-sample distributions used in inference:
 - ▶ normal distributions for linear combinations of normal random variables
 - ▶ chi-square distributions for residual sums of squares
 - ▶ t and F distributions built from those results

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- ▶ It is central for the exact finite-sample distributions used in inference:
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 - ▶ chi-square distributions for residual sums of squares
 - ▶ t and F distributions built from those results
- ▶ Under normal errors, the least-squares estimates are also maximum likelihood estimates.

By the end of this deck, you should be able to:

- ▶ State the PDFs of the univariate and multivariate normal distributions
- ▶ State the MGF of the multivariate normal distribution
- ▶ Use the MGF to prove that a linear transformation of an MVN random vector is MVN
- ▶ Use selection matrices to obtain distributions of components and subvectors
- ▶ Prove the relationship between a standard MVN quadratic form and a χ^2 random variable

Exclusions compared to previous years' courses

The following appear in previous years' slides, tests and some notes, but will not be examined here:

- ▶ Deriving the MVN density
- ▶ Deriving the MVN MGF
- ▶ Determining the mean and covariance matrix of an MVN distribution from its exponent

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The reasons are:

- ▶ Later results can be established without those derivations
- ▶ Learning the derivations well does not substantially help with other course content
- ▶ There is overlap with STA2004F
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You still need to be able to **use** the MGF and the MVN results.

The univariate normal distribution

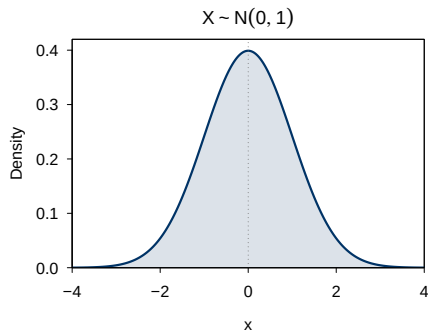
Definition

If $X \sim N(\mu, \sigma^2)$, then its probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad x \in \mathbb{R}.$$

Moreover,

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$



The multivariate normal distribution

Definition

Let $\mathbf{X} = (X_1, \dots, X_p)^T$. If $\mathbf{X} \sim N_p(\mu, \Sigma)$ and Σ is positive definite, then

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right\}, \quad \mathbf{x} \in \mathbb{R}^p.$$

Moreover,

$$E(\mathbf{X}) = \mu, \quad \text{Cov}(\mathbf{X}) = E\{(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T\} = \Sigma.$$

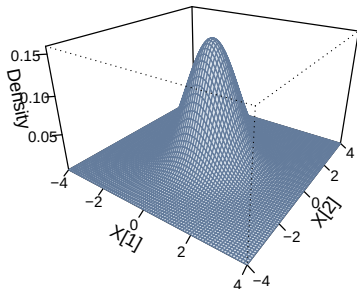
- ▶ Σ is a covariance **matrix**, not a scalar.
- ▶ Its (i, j) entry is $\text{Cov}(X_i, X_j)$.

Visualising a bivariate normal distribution

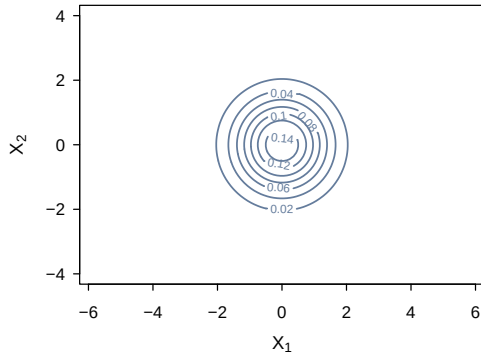
Let

$$\mathbf{X} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right).$$

Density surface



Contour plot



Moment-generating functions

Definition

For a random vector $\mathbf{X}$, the moment-generating function is

$$M_{\mathbf{X}}(\mathbf{t}) = E \{ \exp(\mathbf{t}^T \mathbf{X}) \} ,$$

where $\mathbf{t}$ is a deterministic vector conformable with $\mathbf{X}$.

- ▶ The MGF is said to exist if the expectation is finite in an open neighbourhood of $\mathbf{0}$.
- ▶ For a univariate random variable, derivatives of the MGF at $\mathbf{0}$ give moments.
- ▶ For a random vector, partial derivatives give marginal and mixed moments.

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In this course, the main use of MGFs is not to calculate moments. It is to identify distributions, because an existing MGF uniquely determines the distribution.

The MGF of the multivariate normal distribution

Result

If $\mathbf{X} \sim N_p(\mu, \Sigma)$, then

$$M_{\mathbf{X}}(\mathbf{t}) = \exp \left\{ \mathbf{t}^T \mu + \frac{1}{2} \mathbf{t}^T \Sigma \mathbf{t} \right\}.$$

The MGF of the multivariate normal distribution

Result

If $\mathbf{X} \sim N_p(\mu, \Sigma)$, then

$$M_{\mathbf{X}}(\mathbf{t}) = \exp \left\{ \mathbf{t}^T \mu + \frac{1}{2} \mathbf{t}^T \Sigma \mathbf{t} \right\}.$$

- ▶ We will use this result without deriving it.
- ▶ The important point is that the exponent contains:
 - ▶ a linear term, $\mathbf{t}^T \mu$
 - ▶ a quadratic term, $\frac{1}{2} \mathbf{t}^T \Sigma \mathbf{t}$

Linear transformation of an MVN random variable: theorem

Theorem

Let $\mathbf{X} \sim N_p(\mu, \Sigma)$. If $\mathbf{C}$ is a fixed $q \times p$ matrix, then

$$\mathbf{Y} = \mathbf{CX} \sim N_q(\mathbf{C}\mu, \mathbf{C}\Sigma\mathbf{C}^T).$$

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- ▶ This result is essential because regression estimators are linear transformations of a normally distributed response vector.
- ▶ If $\mathbf{C}\Sigma\mathbf{C}^T$ is singular, the resulting normal distribution is degenerate. The formula for the mean and covariance matrix still holds.

Linear transformation of an MVN random variable: proof

Let $\mathbf{Y} = \mathbf{C}\mathbf{X}$. Then

$$\begin{aligned}M_{\mathbf{Y}}(\mathbf{t}) &= E \{ \exp(\mathbf{t}^T \mathbf{Y}) \} \\&= E \{ \exp(\mathbf{t}^T \mathbf{C}\mathbf{X}) \} \\&= E \{ \exp((\mathbf{C}^T \mathbf{t})^T \mathbf{X}) \} \\&= M_{\mathbf{X}}(\mathbf{C}^T \mathbf{t}).\end{aligned}$$

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Using the MGF of $\mathbf{X}$,

$$\begin{aligned}M_{\mathbf{Y}}(\mathbf{t}) &= \exp \left\{ (\mathbf{C}^T \mathbf{t})^T \boldsymbol{\mu} + \frac{1}{2} (\mathbf{C}^T \mathbf{t})^T \boldsymbol{\Sigma} (\mathbf{C}^T \mathbf{t}) \right\} \\&= \exp \left\{ \mathbf{t}^T \mathbf{C} \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \mathbf{C} \boldsymbol{\Sigma} \mathbf{C}^T \mathbf{t} \right\}.\end{aligned}$$

Linear transformation proof: conclusion

The final expression is the MGF of a multivariate normal random vector with

Identified mean and covariance matrix

$$E(\mathbf{Y}) = \mathbf{C}\mu, \quad \text{Cov}(\mathbf{Y}) = \mathbf{C}\Sigma\mathbf{C}^T.$$

Linear transformation proof: conclusion

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Identified mean and covariance matrix

$$E(\mathbf{Y}) = \mathbf{C}\mu, \quad \text{Cov}(\mathbf{Y}) = \mathbf{C}\Sigma\mathbf{C}^T.$$

Therefore

$$\boxed{\mathbf{Y} = \mathbf{C}\mathbf{X} \sim N_q(\mathbf{C}\mu, \mathbf{C}\Sigma\mathbf{C}^T)}.$$

Why selection matrices are useful

- ▶ A selection matrix is a matrix that extracts components of a vector.
- ▶ This lets us use the linear-transformation theorem to obtain distributions of:
 - ▶ a single component, such as X_1
 - ▶ a subvector, such as $(X_1, X_2)^T$
 - ▶ all components except one, such as $(X_2, \dots, X_p)^T$
 - ▶ a sum, such as $X_1 + \dots + X_p$

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The handwritten examples from the original deck are written out on the next slides.

Example: identity matrix

Suppose

$$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \mathbf{Y} = \mathbf{C}\mathbf{X}.$$

If $\mathbf{C} = \mathbf{I}_p$, then

$$\mathbf{Y} = \mathbf{I}_p \mathbf{X} = \mathbf{X}.$$

Therefore

$$E(\mathbf{Y}) = \mathbf{I}_p \boldsymbol{\mu} = \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{Y}) = \mathbf{I}_p \boldsymbol{\Sigma} \mathbf{I}_p^T = \boldsymbol{\Sigma}.$$

Example: one individual element

Suppose

$$\mathbf{X} = (X_1, \dots, X_p)^T \sim N_p(\mu, \Sigma).$$

To extract X_i , use the row vector

$$\mathbf{e}_i^T = (0 \quad \dots \quad 0 \quad 1 \quad 0 \quad \dots \quad 0),$$

where the 1 is in position i .

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where the 1 is in position i .

Then

$$Y = \mathbf{e}_i^T \mathbf{X} = X_i,$$

so

$$E(Y) = \mathbf{e}_i^T \mu = \mu_i, \quad \text{Var}(Y) = \mathbf{e}_i^T \Sigma \mathbf{e}_i = \sigma_{ii}.$$

Therefore

$$X_i \sim N(\mu_i, \sigma_{ii}).$$

Example: the first two elements

Let

$$\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \end{pmatrix}.$$

Then

$$\mathbf{Y} = \mathbf{CX} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}.$$

Example: the first two elements

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$$\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \end{pmatrix}.$$

Then

$$\mathbf{Y} = \mathbf{CX} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}.$$

By the linear-transformation theorem,

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \right).$$

Example: all elements except the first

Let

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{0}_{(p-1) \times 1} & \mathbf{I}_{p-1} \end{pmatrix}.$$

Then

$$\mathbf{Y} = \mathbf{CX} = \begin{pmatrix} X_2 \\ X_3 \\ \vdots \\ X_p \end{pmatrix}.$$

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Then

$$\mathbf{Y} = \mathbf{C}\mathbf{X} = \begin{pmatrix} X_2 \\ X_3 \\ \vdots \\ X_p \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} X_2 \\ X_3 \\ \vdots \\ X_p \end{pmatrix} \sim N_{p-1} \left(\begin{pmatrix} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_p \end{pmatrix}, \Sigma_{2:p, 2:p} \right).$$

Example: the sum of all elements

Let

$$\mathbf{1}_p = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \quad Y = \mathbf{1}_p^T \mathbf{X}.$$

Then

$$Y = X_1 + X_2 + \cdots + X_p.$$

Example: the sum of all elements

Let

$$\mathbf{1}_p = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \quad Y = \mathbf{1}_p^T \mathbf{X}.$$

Then

$$Y = X_1 + X_2 + \cdots + X_p.$$

Using the linear-transformation theorem,

$$Y \sim N(\mathbf{1}_p^T \mu, \mathbf{1}_p^T \Sigma \mathbf{1}_p).$$

Equivalently,

$$Y \sim N\left(\sum_{i=1}^p \mu_i, \sum_{i=1}^p \sum_{j=1}^p \sigma_{ij}\right).$$

MVN partitions are MVN: theorem

Theorem

Let

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix} \sim N_p \left(\begin{pmatrix} \mu^{(1)} \\ \mu^{(2)} \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right),$$

where $\mathbf{X}^{(1)}$ is $q \times 1$ and $\mathbf{X}^{(2)}$ is $r \times 1$. Then

$$\mathbf{X}^{(1)} \sim N_q(\mu^{(1)}, \Sigma_{11}), \quad \mathbf{X}^{(2)} \sim N_r(\mu^{(2)}, \Sigma_{22}).$$

MVN partitions are MVN: proof

To extract $\mathbf{X}^{(1)}$, define

$$\mathbf{C}_1 = (\mathbf{I}_q \quad \mathbf{0}_{q \times r}).$$

Then

$$\mathbf{C}_1 \mathbf{X} = (\mathbf{I}_q \quad \mathbf{0}) \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix} = \mathbf{X}^{(1)}.$$

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Then

$$\mathbf{C}_1 \mathbf{X} = (\mathbf{I}_q \quad \mathbf{0}) \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix} = \mathbf{X}^{(1)}.$$

The mean is

$$\mathbf{C}_1 \boldsymbol{\mu} = (\mathbf{I}_q \quad \mathbf{0}) \begin{pmatrix} \boldsymbol{\mu}^{(1)} \\ \boldsymbol{\mu}^{(2)} \end{pmatrix} = \boldsymbol{\mu}^{(1)}.$$

MVN partitions are MVN: proof continued

The covariance matrix is

$$\begin{aligned}\mathbf{C}_1 \Sigma \mathbf{C}_1^T &= \begin{pmatrix} \mathbf{I}_q & \mathbf{0} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} \mathbf{I}_q \\ \mathbf{0} \end{pmatrix} \\ &= \Sigma_{11}.\end{aligned}$$

Therefore

$$\mathbf{X}^{(1)} \sim N_q(\mu^{(1)}, \Sigma_{11}).$$

MVN partitions are MVN: proof continued

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Therefore

$$\mathbf{X}^{(1)} \sim N_q(\mu^{(1)}, \Sigma_{11}).$$

The proof for $\mathbf{X}^{(2)}$ is identical, using

$$\mathbf{C}_2 = \begin{pmatrix} \mathbf{0}_{r \times q} & \mathbf{I}_r \end{pmatrix}.$$

Example: bivariate MVN marginals

Suppose

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix} \right).$$

Then, by the partition theorem,

$$X_1 \sim N(\mu_1, \sigma_1^2), \quad X_2 \sim N(\mu_2, \sigma_2^2).$$

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Then, by the partition theorem,

$$X_1 \sim N(\mu_1, \sigma_1^2), \quad X_2 \sim N(\mu_2, \sigma_2^2).$$

- ▶ A jointly normal vector has normal marginal distributions.
- ▶ The converse is not true in general: normal marginal distributions do not by themselves guarantee joint normality.

Quadratic forms and the chi-square distribution

Theorem

Let

$$\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I}_n).$$

If $\mathbf{A}$ is symmetric and idempotent, with rank k , then

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi_k^2.$$

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If $\mathbf{A}$ is symmetric and idempotent, with rank k , then

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi_k^2.$$

- ▶ Idempotent means $\mathbf{A}^2 = \mathbf{A}$.
- ▶ Symmetry is needed for the orthogonal diagonalisation used in the proof.
- ▶ The rank k determines the degrees of freedom.

Quadratic forms and chi-square: proof

Since $\mathbf{A}$ is symmetric and idempotent, there is an orthonormal matrix $\mathbf{P}$ such that

$$\mathbf{P}^T \mathbf{A} \mathbf{P} = \begin{pmatrix} \mathbf{I}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

Let

$$\mathbf{Z} = \mathbf{P}^T \mathbf{Y}.$$

Quadratic forms and chi-square: proof

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Let

$$\mathbf{Z} = \mathbf{P}^T \mathbf{Y}.$$

By the linear-transformation theorem,

$$\mathbf{Z} \sim N_n(\mathbf{P}^T \mathbf{0}, \mathbf{P}^T \mathbf{I}_n \mathbf{P}) = N_n(\mathbf{0}, \mathbf{I}_n),$$

because $\mathbf{P}^T \mathbf{P} = \mathbf{I}_n$.

Quadratic forms and chi-square: proof continued

Since $\mathbf{Y} = \mathbf{PZ}$,

$$\begin{aligned}\mathbf{Y}^T \mathbf{A} \mathbf{Y} &= (\mathbf{PZ})^T \mathbf{A} (\mathbf{PZ}) \\ &= \mathbf{Z}^T \mathbf{P}^T \mathbf{A} \mathbf{P} \mathbf{Z} \\ &= \mathbf{Z}^T \begin{pmatrix} \mathbf{I}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{Z}.\end{aligned}$$

Partition $\mathbf{Z} = (Z_1, \dots, Z_n)^T$. Then

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} = \sum_{i=1}^k Z_i^2.$$

Because $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$, the Z_i are independent standard normal random variables. Hence

$$\boxed{\sum_{i=1}^k Z_i^2 \sim \chi_k^2}.$$

Why this theorem matters for regression

- ▶ In the general linear model, residual sums of squares can be written as quadratic forms.
- ▶ Projection matrices used in regression are symmetric and idempotent.
- ▶ Their ranks determine the relevant degrees of freedom.

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- ▶ In the general linear model, residual sums of squares can be written as quadratic forms.
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- ▶ Their ranks determine the relevant degrees of freedom.

That is why this theorem is one of the foundations for:

- ▶ residual variance estimation
- ▶ t tests for individual coefficients
- ▶ F tests comparing linear models

Summary

In this deck, we used three main ideas:

- ▶ The multivariate normal distribution is stable under linear transformations:

$$\mathbf{CX} \sim N(\mathbf{C}\mu, \mathbf{C}\Sigma\mathbf{C}^T).$$

- ▶ Components and partitions of an MVN random vector are themselves normally distributed.
- ▶ A symmetric idempotent quadratic form in a standard MVN random vector is chi-square:

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2_{\text{rank}(\mathbf{A})}.$$